

# High spatial resolution and large strain measurement in long-distance OFDR

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**Abstract** *We present a distance-domain gating method that simultaneously achieves long sensing distance, high spatial resolution and wide strain measurement range. Strain range of 8000 $\mu\epsilon$  is achieved over a 54m fiber with a 1.9mm spatial resolution. ©2026 The Author(s)*

## Introduction

Distributed optical fiber sensors (DOFSs) can simultaneously measure the spatial and temporal variations of environmental parameters such as strain, temperature, and vibration, which are crucial for numerous applications [1-2]. Among various DOFS techniques, optical frequency domain reflectometry (OFDR) can measure multiple environmental parameters with exceptionally high spatial resolution. The OFDR system employs swept-wavelength homodyne interferometry, an effective approach to overcome the spatial resolution limitations of pulse-modulated systems and to achieve precise localizations along the sensing fiber. To date, OFDR systems have been successfully applied in diverse areas for measuring strain, vibration, temperature, and humidity, as well as for shape sensing, gas detection, and acoustic monitoring [3-6].

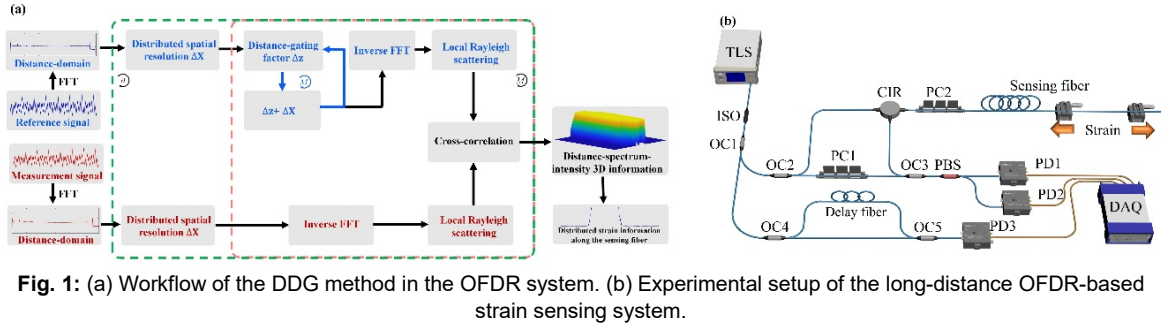
In an OFDR system, spatial resolution, sensing length, and strain measurement range are the primary parameters used to evaluate overall system performance [7-8]. The spatial resolution represents the smallest fiber segment over which environmental changes can be detected, typically corresponding to a region containing tens to hundreds of Rayleigh scattering points. However, random noise sources—such as laser phase noise, polarization fluctuations, and coherence fading often require longer fiber segments to ensure an adequate signal-to-noise ratio (SNR), especially over long sensing distances. This, in turn, reduces spatial resolution and makes it more difficult to reliably extract spectral shift information. The strain measurement range, defined as the detectable strain along the sensing fiber, is generally inversely related to the similarity between the reference and measurement signals at a given spatial resolution. Under large strain conditions, the elongation may become so significant that a wider fiber segment is required, further degrading spatial resolution to accurately capture the spectral shift through cross-correlation. Consequently, trade-offs among spatial resolution, strain range, and sensing length are fundamental performance limits in OFDR systems and developing balanced strategies to explore these trade-

offs is essential for advancing OFDR technology toward intelligent sensing applications.

In this paper, we propose a distance-domain gating method that enables high spatial resolution and a wide strain measurement range in long-distance OFDR systems. A DDG factor is introduced into the reference signal to construct a distance–spectrum–intensity map along the sensing fiber. By applying the proposed method, spatial mismatches caused by large strains and random noise are effectively suppressed, enabling accurate extraction of the true cross-correlation peak. Experimental results demonstrate that a spatial resolution of 1.9 mm and a distributed strain measurement range of up to 8000  $\mu\epsilon$  are simultaneously achieved in an OFDR system with a 54 m sensing length. This technique offers a promising solution for overcoming the challenges of large-strain measurements in distributed optical fiber sensing, with potential applications in soft robotics, intelligent material fatigue testing, and smart building monitoring [9-10].

## Operating Principle of Distance-Domain Gating

The actual spatial resolution attained in OFDR systems is typically lower than the theoretical value, primarily due to the practical challenges in precisely detecting spectral shifts. This resolution is simultaneously affected by the sensing distance and the strain measurement range. In practical measurements, increased sensing distance leads to higher noise levels, which requires a larger number of local Rayleigh backscattering data points ( $N$ ) to ensure reliable correlation, which inevitably leads to a reduction in spatial resolution [11-12]. Additionally, when operating at high spatial resolution, large strain-induced spectral shifts become difficult to accurately identify because of spatial mismatches between the reference and measurement signals [13-14]. Moreover, using high spatial resolution can introduce spatial mismatches beyond the stretched region, further complicating the reliable extraction of variations along the fiber. Consequently,



increasing both the sensing distance and the strain range requires a considerable reduction in spatial resolution to obtain meaningful spectral shift data. This inherent trade-off imposes a critical constraint on OFDR systems, making it challenging to simultaneously achieve long-range, large-strain, and high-spatial-resolution measurements. To overcome the above bottleneck, the conception of DDG is introduced in the OFDR system. The logical flow of the proposed Distance-Domain Gating (DDG) principle for OFDR is illustrated in Fig. 1(a).

**Step 1:** The OFDR system is operated twice to collect two datasets. The first, obtained under non-strain conditions, serves as the reference signal, while the second, recorded under applied strain, serves as the measurement signal.

**Step 2:** By applying the fast Fourier transform (FFT), both the reference and measurement signals are transformed into the distance domain.

**Step 3:** Then, a sliding window,  $\Delta X$ , corresponding to the spatial resolution, is applied to extract the strain information along the fiber.

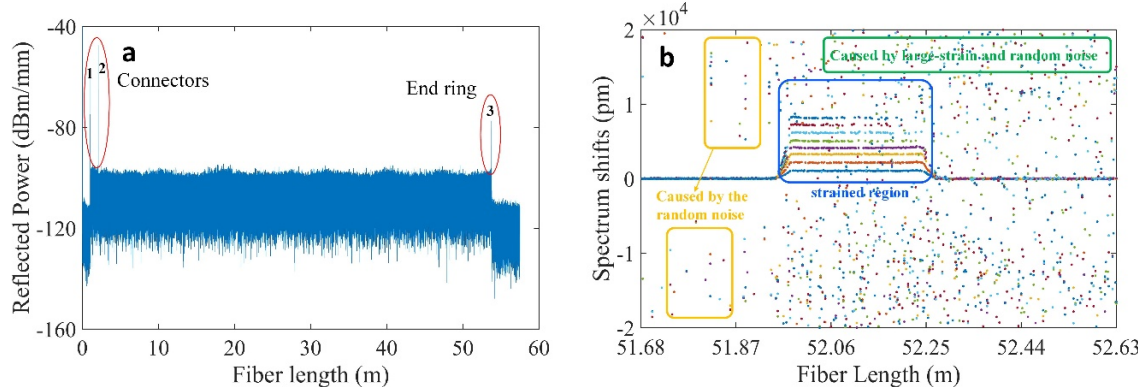
**Step 4:** To compensate for this mismatch and preserve correlation accuracy, the positional shift in the distance domain must be considered. A DDG factor is introduced along the reference signal, defined as:  $\Delta z = \Delta X/M$ . For any fixed position of the measurement signal,  $M$  reference signal segments of equal length but with different distance offsets are sequentially extracted by sliding over the corresponding region.

**Step 5:** These reference signal sets and the

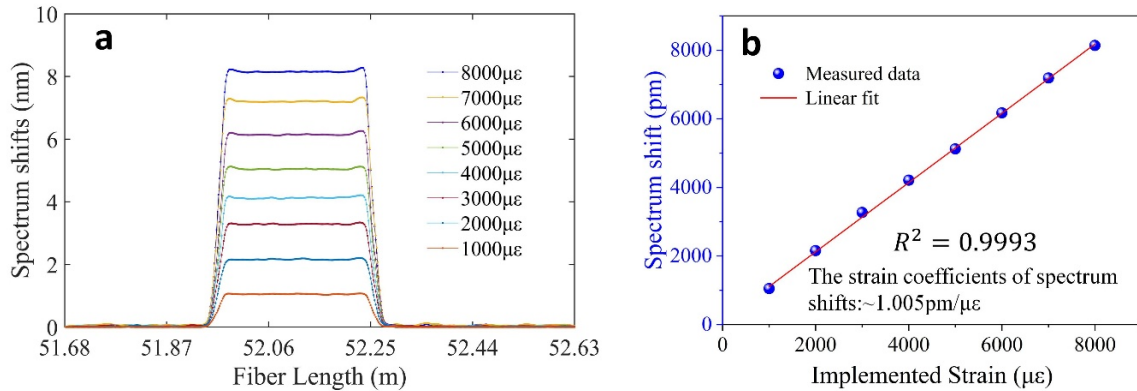
measurement signal are then transformed back to the frequency domain using inverse FFT. Cross-correlation is performed between each reference segment and the measurement signal to construct a set of correlation matrices at that position. These matrices are finally combined to form a 3D distance-spectrum-intensity map from which the distributed strain information along the sensing fiber is precisely recovered.

### Results and Discussions

To verify the effectiveness of the proposed method in enhancing OFDR system performance, a long-distance OFDR-based strain sensing setup is shown in Fig. 1(b). A narrow-linewidth tunable laser source (TLS) is employed as the light source, whose output passes through an optical isolator (ISO) and is subsequently split by a 10/90 optical coupler (OC1). The 90% portion feeds the main interferometer for Rayleigh backscattering detection, while the 10% portion was fed into an auxiliary interferometer which produces an external clock signal. In the main interferometer, the optical signal is further divided by a 1/99 coupler (OC2). The 99% path is launched into the sensing fiber, while the 1% path passed through PC1 to align the polarization in the local reference arm. The beat signal is formed at a 50/50 coupler (OC3). A polarization beam splitter (PBS) then separates the beat signal into 'p' and 's' channels, which are detected by a balanced photodetector (PD) and digitized by a data acquisition (DAQ) card. Meanwhile, the auxiliary interferometer, configured as an imbalanced Mach-Zehnder interferometer consisting of two



**Fig. 2:** (a) The distance domain information of the reference signal in the OFDR system. (b) spectral shifts along the sensing fiber under different applied strains with spatial resolution of 1.9mm by using the traditional method.



**Fig. 3:** (a) Spectrum shift distributions along the fiber under different applied strains ranging from 1000μϵ to 8000μϵ. (b) Relationship between the implemented strain and the corresponding spectrum shift.

50/50 couplers (OC4, OC5) and a delay fiber, produces a stable clock signal. The interference output from OC5, detected by PD3, serves as the timing trigger for the DAQ, maintaining synchronization between the wavelength sweep and data sampling.

In this system, the TLS was linearly swept from 1520nm to 1570nm at a rate of 10nm/s. The optical path difference in the auxiliary interferometer was set to 230m, and the probe light was launched into a 54m standard single-mode fiber through a circulator. The distance-domain data, shown in Fig. 2(a), were obtained by vector summing the “s” and “p” polarization components, representing Rayleigh backscattering intensity as a function of fiber length.

After configuring the experimental parameters, two fiber-optic micro-displacement platforms positioned near the 52m location were used to apply controlled strain to a 30cm segment at the distal end of the sensing fiber. Strain levels ranging from 1000μϵ to 8000μϵ, in increments of 1000μϵ, were applied and sequentially recorded using a data acquisition (DAQ) card. A sliding window was adjusted to include 100 Rayleigh scattering data points, corresponding to a spatial resolution of 1.9mm. The strain distribution obtained using the conventional approach is shown in Fig. 2(b). As observed, the fiber segment near the 52m position experienced controlled tensile strain ranging from 1000μϵ to 8000μϵ. Within the strained region, the spectral shifts display a step-like profile, with their magnitude increasing proportionally to the applied strain. The outliers distributed along the fiber result from the combined influence of system noise and reduced signal similarity between the reference and measurement traces under large strain. In contrast, spurious spectral spikes appear outside the strained region. These spurious responses hinder accurate identification of the fiber state and highlight the limitations of conventional processing methods under large-strain conditions.

Fig. 3(a) presents the spectral shift distribution along the sensing fiber under various applied strain levels, measured with a spatial resolution of 1.9 mm using the proposed method. The spectral response remains unchanged in unstrained regions, while a distinct shift appears exclusively within the strained segment, allowing for straightforward identification of the strained zone. In contrast to the conventional method, the proposed approach effectively suppresses false peaks caused by random noise and large strain variations along the fiber. Moreover, it eliminates spurious peaks outside the strained region, ensuring accurate localization of the applied strain. These results confirm that the proposed method maintains high spatial resolution, enabling precise localization of the strained segment and reliable tracking of spectral variations induced by large strain in a long-distance OFDR system. Fig. 3(b) shows the linear relationship between the applied strain and the corresponding spectral shifts. The strain sensitivity coefficient and coefficient of determination ( $R^2$ ) are approximately 1.005 pm/μϵ and 0.9993, respectively. These results demonstrate a clear and well-defined linear correlation between the applied strain and the corresponding spectral shift.

### Conclusions

We proposed Distance-Domain Gating to enhance the measurement performance of OFDR sensing systems and it has been both theoretically formulated and experimentally validated for distributed strain measurement. Compared with conventional approaches, the proposed method eliminates erroneous “ghost points” caused by reduced signal similarity from strain-induced stretching and random noise, while simultaneously achieving long-distance sensing, high spatial resolution, and a wide strain measurement range.

## Acknowledgements

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## References

- [1] N. J. Lindsey, T. C. Dawe, and J. B. Ajo-Franklin, "Illuminating seafloor faults and ocean dynamics with dark fiber distributed acoustic sensing," *Science*, vol. 366, no. 6469, pp. 1103-1107, 2019, DOI: [10.1126/science.aay5881](https://doi.org/10.1126/science.aay5881).
- [2] G. Marra, D. M. Fairweather, V. Kamalov, et al., "Optical interferometry-based array of seafloor environmental sensors using a transoceanic submarine cable," *Science*, vol. 376, pp. 874-879, 2022, DOI: [10.1126/science.abo1939](https://doi.org/10.1126/science.abo1939).
- [3] Z. Qin, S. Qu, Z. Wang, et al., "A fully distributed fiber optic sensor for simultaneous relative humidity and temperature measurement with polyimide-coated polarization maintaining fiber," *Sens. Actuators B, Chem.*, vol. 373, pp. 132699, 2022, DOI: [10.1016/j.snb.2022.132699](https://doi.org/10.1016/j.snb.2022.132699).
- [4] X. Lou, Y. Wang, N. Xu, and Y. Dong, "Gas sensing with 7-decade dynamic range by laser vector spectroscopy combining absorption and dispersion," *Photonics Research*, vol. 11, pp. 1687-1693, 2023, DOI: [10.1364/PRJ.492651](https://doi.org/10.1364/PRJ.492651).
- [5] C. Fu, S. Xiao, Y. Meng, et al., "OFDR shape sensor based on a femtosecond-laser-inscribed weak fiber Bragg grating array in a multicore fiber," *Optics Letters*, vol. 49, pp. 1273-1276, 2024, DOI: [10.1364/OL.516067](https://doi.org/10.1364/OL.516067).
- [6] Y. Yu, E. Vergori, D. Worwood, et al., "Distributed thermal monitoring of lithium ion batteries with optical fibre sensors," *Journal of Energy Storage*, vol. 39, pp. 102560, 2021, DOI: [10.1016/j.est.2021.102560](https://doi.org/10.1016/j.est.2021.102560).
- [7] J. P. von der Weid, R. Passy, G. Mussi, and N. Gisin, "On the characterization of optical fiber network components with optical frequency domain reflectometry," *Journal of Lightwave Technology*, vol. 15, no. 7, pp. 1131-1141, 1997, DOI: [10.1109/50.596958](https://doi.org/10.1109/50.596958).
- [8] Z. Ding, X. S. Yao, T. Liu, et al., "Long-range vibration sensor based on correlation analysis of optical frequency-domain reflectometry signals," *Optics Express*, vol. 20, pp. 28319-28329, 2012, DOI: [10.1364/OE.20.028319](https://doi.org/10.1364/OE.20.028319).
- [9] Y. Yin, P. Yu, Z. Jia, et al., "An approach of the maximum curvature measurement of dynamic umbilicals using OFDR technology in fatigue tests," *Frontiers in Materials*, vol. 8, pp. 717190, 2021, DOI: [10.3389/fmats.2021.717190](https://doi.org/10.3389/fmats.2021.717190).
- [10] H. Bai, S. Li, J. Barreiros, Y. Tu, C. Pollock, and R. Shepherd, "Stretchable distributed fiber-optic sensors," *Science*, vol. 370, no. 6518, pp. 848-852, 2020, DOI: [10.1126/science.aba5504](https://doi.org/10.1126/science.aba5504).
- [11] S. Qu, Z. Qin, Y. Xu, et al., "High spatial resolution investigation of OFDR based on image denoising methods," *IEEE Sensors Journal*, vol. 21, pp. 18871-18876, 2021, DOI: [10.1109/JSEN.2021.3089479](https://doi.org/10.1109/JSEN.2021.3089479).
- [12] S. Zhao, J. Cui, Z. Wu, et al., "Accuracy improvement in OFDR-based distributed sensing system by image processing," *Optics and Lasers in Engineering*, vol. 124, pp. 105824, 2020, DOI: [10.1016/j.optlaseng.2019.105824](https://doi.org/10.1016/j.optlaseng.2019.105824).
- [13] S. Zhao, J. Cui, L. Suo, et al., "Performance investigation of OFDR sensing system with a wide strain measurement range," *Journal of Lightwave Technology*, vol. 37, pp. 3721-3727, 2019, DOI: [10.1109/JLT.2019.2918379](https://doi.org/10.1109/JLT.2019.2918379).
- [14] K. Feng, J. Cui, D. Jiang, et al., "Improvement of the strain measurable range of an OFDR based on local similar characteristics of a Rayleigh scattering spectrum," *Optics letters*, vol. 43, pp. 3293-3296, 2018, DOI: [10.1364/OL.43.003293](https://doi.org/10.1364/OL.43.003293).